



SIKSHA 'O' ANUSANDHAN

(Deemed to be University)

Faculty of Engineering & Technology (ITER)

Department of Computer Science and Engineering

Project Proposal Form

FINAL YEAR RESEARCH PROJECT-2026

SECTION: 04

GROUP NO: 09

PROJECT TITLE: Development and Comparative Evaluation of a Cost-Aware Ensemble Machine Learning Framework for Anaemia Prediction Using DHS/NFHS Data

PROJECT ABSTRACT:

Anaemia remains a major public health concern in developing regions, particularly in low-resource settings where access to advanced diagnostic infrastructure may be limited. This project aims to develop and comparatively evaluate a cost-aware ensemble machine learning framework for anaemia prediction using demographic and health-related variables derived from Demographic and Health Survey (DHS) and National Family Health Survey (NFHS) datasets. The study focuses on designing predictive models that maintain strong performance while considering feasibility in resource-constrained public health environments.

Initially, baseline machine learning models such as Logistic Regression, Random Forest, Artificial Neural Networks, and gradient boosting techniques will be implemented. An existing hybrid stacking architecture proposed in prior literature will be replicated under identical experimental conditions to ensure fair comparison. Subsequently, an enhanced ensemble stacking framework will be developed by integrating additional boosting strategies to investigate whether improved predictive performance can be achieved.

Model evaluation will be conducted using standardized metrics including accuracy, Area Under the Curve (AUC), sensitivity, and specificity, along with cross-validation to ensure robustness. Furthermore, anemia prediction performance will be examined with specific reference to Odisha using NFHS data to explore regional applicability and public health relevance.

The overall objective is to assess whether advanced ensemble learning techniques can provide meaningful improvements in anaemia prediction accuracy while remaining adaptable for use in large-scale population health contexts.

(1) SOFTWARE, HARDWARE OR METHODS/ALGORITHMS SPECIFICATIONS:

Software Specifications

- Programming Language: Python 3.13
- Development Environment: Visual Studio Code
- Libraries Used:
 - NumPy – Numerical computation and matrix operations
 - Pandas – Data preprocessing, cleaning, and manipulation
 - Scikit-learn – Machine learning algorithms, model training, cross-validation, and evaluation
 - XGBoost – Gradient boosting classifier for baseline and meta-learning experiments
 - LightGBM – Boosting-based ensemble learning (enhanced stacking framework)
 - CatBoost – Gradient boosting with categorical handling (meta-learner for stacking experiments)
 - Matplotlib / Seaborn – Data visualization and performance analysis
 - Statsmodels (if required) – Statistical validation and hypothesis testing
- Operating System: Windows 11

Hardware Specifications

- Device: HP Victus Laptop
- Processor: AMD Ryzen Series Processor (Multi-core CPU architecture)
- RAM: 16 GB DDR4
- Storage: SSD-based system
- GPU: Not required (CPU-based model training and evaluation performed)

Methods / Algorithms Used

The project focuses on anaemia prediction using supervised machine learning techniques and ensemble stacking architectures applied to DHS/NFHS population health data.

Baseline Models

The following individual classifiers will be implemented as baseline models:

- Logistic Regression
- Random Forest Classifier
- Artificial Neural Network (MLP Classifier)
- XGBoost Classifier

Hybrid Stacking Architecture (Replication Phase)

To ensure methodological comparison with prior literature, a hybrid stacking model inspired by the Tanzania study will be implemented.

- Base Learners:
 - Random Forest
 - Artificial Neural Network
- Meta Learner:
 - XGBoost

This architecture will be trained and evaluated under identical experimental conditions.

Proposed Improved Ensemble Framework

An enhanced stacking architecture will be developed to investigate performance improvements.

- Base Learners:
 - Random Forest
 - Artificial Neural Network
 - LightGBM
- Meta Learner:
 - CatBoost

The improved architecture aims to enhance generalization performance and robustness.

Evaluation Metrics

Model performance will be assessed using:

- Accuracy
- Area Under Curve (AUC)
- Sensitivity (Recall)
- Specificity
- Confusion Matrix Analysis

Stratified train-test splitting and K-fold cross-validation will be applied to ensure robust evaluation.

Data Processing Approach

- Dataset loading and preprocessing using Pandas
- Feature encoding and normalization where required
- WHO-based anemia label construction
- Handling missing values and outliers
- Stratified Train-Test Split (70:30 ratio)
- Probability-based stacking for ensemble learning
- Comparative performance analysis across baseline and ensemble models

(2) NAME, REG. NO AND SIGNATURE OF GROUP MEMBERS:

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(3) APPROVAL STATUS(To be filled in by the Section Coordinator of FRP):

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Project Supervisor

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Section Coordinator, FRP

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FRP Coordinator